

GROUP 6 | MODULE: EC6204

Improving DES Security Through **Pseudo-Random** Key Schedule Generation

Information Security Mini Project Proposal
University of Ruhuna, Faculty of Engineering

The Problem Statement

- ⚠ **Predictable Pattern:** The DES key schedule uses a fixed, publicly known shift schedule that is completely independent of the secret key.
- ⚠ **Weak Diffusion:** A single-bit change in the master key does not propagate adequately across all 16 subkeys.
- ⚠ **Vulnerability:** This predictability exposes the cipher to partial subkey recovery attacks.
- ⚠ **Reduced Complexity:** Effective cryptanalysis complexity is drastically reduced well below the theoretical bound of 2^{56}

Proposed Modification



LCG Integration

Replace the fixed shift schedule with a lightweight Linear Congruential Generator (LCG). This generator will be uniquely seeded with the 56-bit DES master key.



Dynamic Shifts

The LCG produces a highly unique sequence of shift amounts (1 or 2) for each of the 16 encryption rounds, ensuring dynamic subkey generation.



Preserved Logic

The cumulative shift total is preserved to maintain correctness. Only the key schedule is modified; all core encryption and decryption logic remains unchanged.

Objectives & Scope

Project Objectives

- ✔ Document the inherent security weakness of the original fixed shift schedule.
- ✔ Design a robust key-dependent pseudo-random shift schedule for subkeys.
- ✔ Implement original and modified key schedules in Python using modular code.
- ✔ Demonstrate measurable improvements via avalanche effect and entropy.
- ✔ Verify 100% decryption correctness across all cipher test cases.

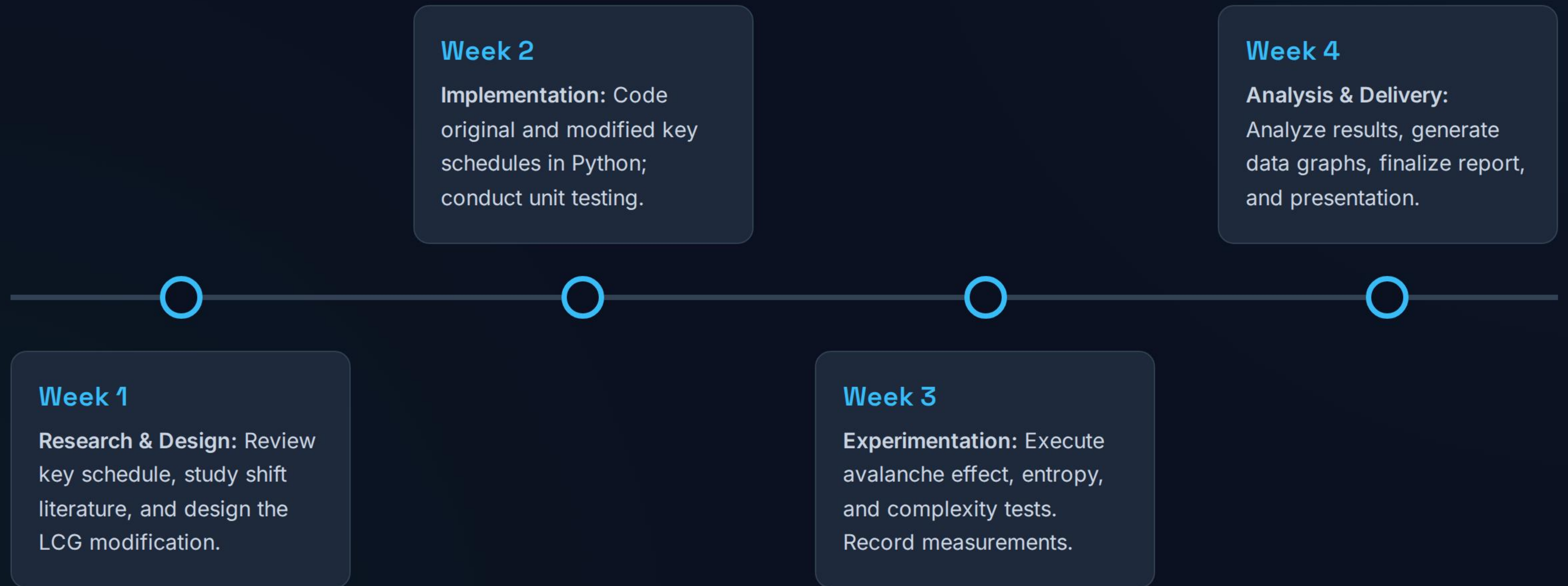
Defined Scope

- ✘ Focus is strictly limited to the key schedule component.
- ✘ Not attempting a full cipher redesign or structural overhaul.
- ✘ Tested entirely on randomly generated 56-bit keys and 64-bit plaintext blocks.
- ✘ Addressing S-box design flaws is explicitly out of scope.
- ✘ Addressing the 56-bit key size limitation is out of scope.

Evaluation Metrics Comparison

Metric	Original DES (Baseline)	Modified DES (Expected)
Avalanche Effect	Fixed, highly predictable pattern	Higher bit-change % per subkey
Subkey Entropy	Low — schedule is deterministic	Increased Shannon entropy
Brute-force Complexity	Below theoretical limit due to pattern	Closer to theoretical 2^{56}
Correctness	100% decryption accuracy	Must remain 100% accurate
Computational Overhead	Baseline metric	Negligible PRNG overhead

Project Timeline



Questions?

Thank you for your time and attention.

Image Sources



https://images.stockcake.com/public/7/a/5/7a5cb724-9864-4235-b2ab-1ee0f0080700_large/digital-security-shield-stockcake.jpg

Source: stockcake.com